



Human Activity Recognition (HAR) Project | Phases 1 & 2 Report

Course: Supervised Learning

Project ID: 6

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1. Project Objective

The goal of this project is to develop a robust supervised learning pipeline capable of classifying human physical activities (Walking, Walking_Upstairs, Walking_Downstairs, Sitting, Standing, Laying) using tri-axial signal data from smartphone accelerometers and gyroscopes. This report covers the foundational data engineering (Phase 1) and the analytical modeling (Phase 2).

2. Phase 1: Data Engineering & Foundation

Focus: Data Acquisition, Cleaning, and Structural Preparation.

In the initial phase, the UCI HAR dataset was downloaded and programmatically extracted. Key technical steps included:

- **Data Loading:** Mapping the 561-feature vector from raw text files (`X_train.txt`) and aligning them with their respective activity labels (`y_train.txt`).
- **Metadata Mapping:** Utilizing `features.txt` to assign descriptive column names to the dataset, ensuring the data is interpretable for later analysis.
- **Serialization:** To optimize Phase 2, the data was cleaned of potential null values and exported to `train_features.csv` and `test_features.csv`, establishing a stable starting point for modeling.

3. Phase 2: Intelligent Data Analysis (IDA) & Modeling

Focus: Exploratory Analysis and Supervised Model Benchmarking.

3.1 Intelligent Data Analysis (IDA)

Using `ida.py`, we performed a deep dive into the sensor signals:

- **Signal Patterns:** Visualized the first three features (X, Y, Z axes), identifying clear signature differences between dynamic movements and static positions.
- **Feature Correlation:** Generated a heatmap for the first 20 features, highlighting strong dependencies between sensor axes, which informs future dimensionality reduction needs.
- **Statistical Distribution:** Confirmed normalized distributions for several features through histogram and KDE plots, satisfying the assumptions for probabilistic models.

3.2 Supervised Learning Implementation

The core logic in `ml_modes.py` evaluates several classification paradigms:

Model Type	Implementation	Role in Project
Instance-Based	K-Nearest Neighbors (KNN)	Baseline for distance-based similarity in high dimensions.
Probabilistic	Gaussian Naive Bayes	Tests the efficiency of Gaussian distribution assumptions.
Ensemble (Bagging)	Random Forest	Handles non-linear relationships and reduces overfitting.
Ensemble (Boosting)	Gradient Boosting	Optimized for high accuracy through sequential error reduction.

Directory/ ├── project.ipynb # Integrated Research (Phases 1 & 2 combined) ├── ida.py # Exploratory Visualizations & Statistical Checks ├── ml_modes.py # Machine Learning Pipeline(training/Evaluation) ├── train_features.csv # Structured Training Data ├── test_features.csv # Structured Testing Data

4. Phase 3: Advanced Deep Learning & Generative Modeling

Focus: Temporal Pattern Capturing and Data Synthesis. While Phase 2 focused on classical feature-based models, Phase 3 transitions to raw inertial signals to capture complex temporal dependencies using Deep Learning architectures.

4.1 Sequential Modeling: CNN, LSTM, and Transformers

We implemented several neural network architectures to evaluate their ability to classify activities from raw 128-sample signal windows:

- **Convolutional Neural Networks (CNN):** Utilized Conv1D layers and MaxPooling to extract local spatial-temporal features from the sensor channels.
- **Long Short-Term Memory (LSTM):** Designed to capture long-range temporal dependencies in the movement data. We also evaluated a Bidirectional LSTM (BiLSTM) to process sequences in both directions.
- **Attention Mechanism:** An LSTM model was augmented with an Attention layer to highlight specific time steps within a signal window that are most critical for classification.
- **Transformer Encoder:** Implemented a multi-head attention-based encoder to test the efficiency of parallelized sequential processing.

Performance Comparison Table: The following metrics represent the final evaluation on the raw inertial test set:

Model Type	Accuracy	F1 Score (Weighted)	Parameters
CNN	0.9155	0.9154	169,542
Transformer Encoder	0.9053	0.9041	72,134
LSTM	0.9050	0.9050	124,614
BiLSTM	0.8941	0.8938	314,310
LSTM + Attention	0.8935	0.8941	87,558

4.2 Generative Modeling: Conditional GAN

To address potential data scarcity and explore data synthesis, we developed a Conditional Generative Adversarial Network (cGAN).

- **Generator:** Learns to produce synthetic 1D sensor signals (accelerometer/gyroscope) conditioned on a specific activity label.
- **Discriminator:** Trained to distinguish between real UCI HAR signals and synthetic samples.
- **Evaluation:** The cGAN successfully generated samples that mimic the statistical properties of dynamic activities like "Walking" and static positions like "Laying".

4.3 Explainable AI (XAI) with SHAP

To make the "black box" models interpretable, we applied SHAP (SHapley Additive exPlanations). This allowed us to quantify the contribution of different sensor channels to the model's decisions.

Key XAI Insight: Feature Importance by Sensor Channel

Sensor Channel	Total Mean SHAP value
Total Acc X	0.0256
Total Acc Y	0.0288
Body Gyro X	0.0086
Body Acc X	0.0080

5. Final Conclusions

The transition from classical machine learning to Deep Learning architectures in Phase 3 provided a more direct way to process raw signals without extensive manual feature engineering. The CNN model emerged as the top performer with an accuracy of 91.55%, proving highly effective at capturing signature patterns in tri-axial signal data. The inclusion of XAI confirmed that global acceleration (Total Acc) remains the most critical factor in distinguishing between different human physical activities.